ANKIT **KUMAR**

Blockchain Developer

Edinburgh, UK

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“Embrace complexity, drive innovation, and rewrite the future – one block at a time.“

Summary

Results-driven Blockchain Developer with 5 years of experience specializing in Cosmos-SDK and Tendermint-based ecosystems. Expertise in designing decentralized systems, implementing Inter-Blockchain Communication (IBC), and integrating Data Availability layers like Celestia. Proficient in Rust, Golang, Solidity, and CosmWasm for smart contract and module development. Skilled in building scalable backend services using Python (Django), Node.js, and managing DevOps pipelines with Docker. Experienced with Sui, Ethereum, Quorum, and Hyperledger protocols, and adept at developing tokenization, NFT, and DeFi solutions. Proven success in delivering production-grade blockchain projects, optimizing performance, and ensuring security. Strong collaboration skills with remote and distributed teams, leveraging tools like Slack and JIRA to coordinate across time zones. Seeking remote opportunities to build innovative blockchain applications in global markets.

Work Experience

ENODA — *Edinburgh. UK*

Blockchain Engineer — *July. 2025 - current*

- Defined technical standards and authored 30+ documents shaping ENSEMBLE’s internal knowledge base by analyzing Sui protocol behavior, documenting ENSEMBLE (Sui) network operations and mentored junior engineers for benchmarking validator throughput.
- Owned the development of ENSEMBLE’s monitoring stack using Prometheus, Loki, and Tempo, enabling faster node diagnosis through a unified Grafana dashboard.
- Contributed to oracle research for fleet telemetry, designing a custom telemetry-to-oracle pipeline.
- Independently discovered and analyzed a critical consensus-level bug in a PoC ENSEMBLE (Cosmos SDK) implementation, causing non-deterministic state transitions.
- Deployed a devnet for ENSEMBLE (Cosmos-SDK) network with integrated Prometheus & Loki monitoring and a faucet, enabling up to 10K TPS.
- Owned infrastructure automation for ENSEMBLE and Conductor, reducing cloud costs by 30% and clearing a three-month critical blocker.

Recro, (Zeeve) — *Remote*

Blockchain Research Lead — *July. 2024 - June. 2025*

- Led a team of 3 engineers in building the Cero chain (Cosmos SDK-based rollup sequencing network), including architecture, development, and a CometBFT (Tendermint) consensus upgrade tailored to its requirements.
- Defined core infrastructure and implemented ABCI communication for seamless data sharing between application and consensus layers.
- Integrated Celestia Data Availability with Cero and developed a custom relayer for interoperability.
- Independently researched and implemented a custom gas model (EIP-1559) for Optimistic Rollups on the application layer using op-geth.

C3ihub, IIT Kanpur — *Kanpur, India*

Blockchain Developer — *Feb. 2023 - July. 2024*

- Conceptualized and implemented W3C-compliant DIDs using Solidity, with comprehensive APIs for seamless integration.
- Developed a React Native application for secure medical record sharing, implementing TEE (Hyperledger-based system handling 100 TPS).
- Tokenized real estate RWA for the KDA pilot project, focusing on on-chain development rights management using Quorum.
- Directed the development of four key services, including payment gateways and user registration systems, using Django DRF, MongoDB, & Redis.
- Led a team of 2 engineers to architect a blockchain-based e-voting platform for university alumni, reducing API response time by 40% and enabling seamless handling of 10,000+ concurrent users (Polygon-based).
- Emphasized CI/CD pipelines, security, and performance optimizations throughout 4+ projects.

DotSei — *Remote*

Freelance Blockchain Developer — *May 2023 - Aug. 2023*

- Led the development of a naming service, equipped with an optimized resolver architecture, focusing on cross-chain interoperability.
- Designed and deployed CosmWasm-based contracts (Registry, Name Service Contract, Resolver) using Rust.

AssetMantle (Cambrian Technologies Private Company) — *Bengaluru, India*

Blockchain Developer — *March 2022 - Jan. 2023*

- Developed NFT modules for the AssetMantle chain within the Cosmos ecosystem, focusing on IBC compatibility and module scalability.
- Managed wallet (React), blockchain explorer (Scala/Play Framework), and airdrop implementations.

Persistence One, (CMMT TECHNOLOGIES PVT. LTD) — *Bengaluru, India*

Blockchain Developer — *July 2021 - Feb. 2022*

- Worked on PersistenceCore, enhancing the Persistence chain’s Cosmos-SDK-based infrastructure.
- Improved the blockchain explorer’s performance by optimizing data processing pipelines.
- Collaborated on integrating gRPC APIs for client interactions with Persistence nodes.

Certifications

UDACITY

BLOCKCHAIN NANODEGREE PROGRAM *May 2021 - June 2021*

- Learnt about Blockchain fundamentals, Ethereum smart contracts, Tokens etc.
- Developed DApps with autonomous smart contracts, implemented on the Rinkeby Ethereum test network.

Interchain Foundation & B9Lab

INTERCHAIN DEVELOPER ACADEMY - CERTIFICATION *Sept 2022 - Dec 2022*

- Acquired in-depth knowledge on blockchain fundamentals, smart contracts, and development.
- Gained hands-on experience in building and deploying decentralized applications.
- Learnt cosmos module development, working with protobuf gRPC & IBC, relayers, cosmos-js, ignite-cli, cosmwasm, using ignite-cli and without using ignite-cli
- Developed a module for a toll road, implementing road operator and user logics on a cosmos-based chain.
- Set up validator, relayer, and worked on chain upgrades.

Udemy (Instructor: Daniel Ciocîrlan)

AKKA REMOTING AND CLUSTERING WITH SCALA | ROCK THE JVM *Oct - 2021*

- Acquired in-depth knowledge of Akka remoting, clustering, and Scala programming.
- Gained hands-on experience in building distributed systems with Akka toolkit.

Udemy (Instructor: Stephen Grider)

GO: THE COMPLETE DEVELOPER'S GUIDE (GOLANG) *Feb - 2021*

- Mastered the fundamentals of the Go programming language including concurrency, Go routines, and interfaces.
- Developed efficient, type-safe programs and APIs with Go.

Udemy (Instructor: Daniel Ciocîrlan)

ADVANCED SCALA AND FUNCTIONAL PROGRAMMING | ROCK THE JVM *July - 2021*

- Enhanced understanding of Scala, including advanced functional programming concepts.
- Built complex and scalable applications using Scala's powerful features.

Education

NIT Silchar, Assam

Assam, India

B.TECH, ELECTRICAL ENGINEERING *2017 - 2021*

- Studied core electrical engineering subjects along with computer science electives.
- Worked on several projects related to blockchain technology.

Skills

Blockchain Development	Mysticeti, Solidity, Golang (Cosmos-SDK), CosmWasm (Rust), IBC (Inter-Blockchain Communication), Optimistic Rollups, Tendermint (CometBFT), BFT Consensus, Bitcoin Scripts
Blockchain Protocols & Tools	Sui, Cosmos, Ethereum, Bitcoin, Quorum, Polygon, Hyperledger Fabric, Celestia, Truffle, web3.js, cosmos-js
Backend Development	Scala (Play Framework), Python (Django), Node.js (Express.js), REST APIs, gRPC APIs, Postgres, MongoDB, Redis, IPFS
DevOps & Deployment	AWS EC2, RDS, Docker, Ansible, GitHub Actions (CI/CD), Prometheus, Loki, Tempo, Grafana, GCP
Frontend & App Development	React Native, React, Redux, React Native Keychain
Security & Testing	Smart Contract Auditing (MythX, Slither), Test-Driven Development, Keystore Management
Programming Languages	Rust, Go, Solidity, JavaScript/TypeScript, Python, Scala, Bash
Languages	English, Hindi

Projects

Hobby

HARDWARE BASED

- Designed and developed a Digispark (Arduino)-based Rubber Ducky, capable of executing scripted payloads by emulating a keyboard. This cost-effective device provides a 70% price reduction compared to commercially available alternatives.
- Created an Arduino Uno-based NFC card reader and writer, offering a budget-friendly alternative to the Proxmark3 at a 50% lower cost.
- Conducted controlled wireless security testing on Wi-Fi networks in a controlled home lab environment, including packet capturing, deauthentication attacks, and denial-of-service (DoS) attacks.
- Gained proficiency in packet capturing using HackRF for Software-Defined Radio (SDR).